

CyberSOC Platform

Billing & Licensing Architecture

Enterprise-Grade Multi-Tier Licensing Framework
for AI-Native Security Operations Center

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1. Executive Summary

This document presents a comprehensive Billing and Licensing Architecture designed specifically for the CyberSOC Platform, an AI-Native Security Operations Center operating system. The architecture addresses the complex requirements of modern cybersecurity platforms that must serve diverse customer segments ranging from small security teams to large enterprise organizations and government agencies. The multi-tier approach ensures scalability, flexibility, and compliance across all deployment scenarios while maintaining a unified and manageable licensing framework that can adapt to evolving market demands.

The CyberSOC licensing model introduces six distinct tiers: Free, Professional, Business, Enterprise, Government, and On-Premise. Each tier is carefully calibrated to match specific organizational needs, budget constraints, and operational requirements. The Free tier serves as an entry point for evaluation and small-scale deployments, while the Enterprise and Government tiers provide advanced features necessary for large-scale security operations with stringent compliance requirements. The On-Premise option addresses the critical need for air-gapped environments where cloud connectivity is restricted or prohibited by security policy or regulatory mandate.

Key architectural principles underlying this design include modular feature packaging that allows customers to select precisely the capabilities they need without paying for unnecessary functionality. The usage-based pricing components ensure fairness and predictability, while the volume-based discount structures reward customer loyalty and growth. The license enforcement mechanism balances robustness with user experience, preventing revenue leakage while minimizing friction for legitimate users. Integration with the existing CyberSOC Admin Interface ensures seamless management of all licensing operations through a unified control plane.

This specification serves as the authoritative reference for engineering teams implementing the billing system, product managers defining packaging strategies, sales teams configuring quotes, and customer success teams managing upgrades and renewals. All stakeholders should reference this document when making decisions related to pricing, packaging, license enforcement, or subscription lifecycle management. The architecture has been designed with extensibility in mind, allowing new tiers, features, or pricing models to be introduced without disrupting existing customer deployments or requiring fundamental changes to the core licensing infrastructure.

Architecture Metrics			
Total Tiers	6	Feature Categories	12
Price Points	6 Base + Usage	Deployment Models	Cloud + On-Prem
Integration Points	8 Modules	Compliance Standards	5 Frameworks

Table 1.1: Architecture Overview Metrics

2. Licensing Tiers Overview

The CyberSOC Platform employs a six-tier licensing model that spans the full spectrum of organizational needs from individual security practitioners to large government agencies. Each tier represents a distinct value proposition with carefully curated feature sets, support levels, and deployment options. The tier structure follows industry best practices while incorporating innovations specific to AI-native security platforms. Understanding the positioning and target audience for each tier is essential for effective sales execution, customer success management, and product roadmap planning.

2.1 Tier Definitions and Target Segments

Free Tier (Community Edition): The Free tier provides foundational security capabilities at no cost, designed primarily for educational purposes, individual researchers, small teams evaluating the platform, and open-source community engagement. This tier includes basic SIEM functionality with limited data retention (7 days), up to 5 users, 1 GB/day log ingestion, and community forum support. The Free tier serves as the primary acquisition channel, allowing potential customers to experience core platform value before committing to a paid subscription. Strategic importance extends beyond direct revenue to include market awareness, developer ecosystem building, and competitive displacement of legacy SIEM solutions in small deployments.

Professional Tier: Targeted at small to medium security teams (10-50 analysts), the Professional tier adds advanced detection capabilities, extended data retention (30 days), increased ingestion capacity (10 GB/day), and email support with 4-hour response SLA. This tier includes UEBA basics, SOAR workflow automation (50 actions/month), and API access for custom integrations. The Professional tier represents the primary conversion path from Free users who have validated platform fit and require production-grade capabilities. Pricing is structured as per-analyst-per-month to align with team scaling patterns typical of growing security operations functions.

Business Tier: Designed for established security operations centers (50-200 analysts), the Business tier introduces collaborative features essential for team coordination at scale. Key additions include 90-day data retention, 100 GB/day ingestion, XDR endpoint integration, threat intelligence feeds, case management with full DFIR workflow, and dedicated success manager. Support escalates to priority phone support with 2-hour response SLA and quarterly business reviews. The Business tier often serves as the entry point for enterprises beginning their CyberSOC journey, providing sufficient capability for initial deployment while establishing foundation for future expansion to Enterprise tier requirements.

Enterprise Tier: The flagship offering for large organizations (200+ analysts), the Enterprise tier unlocks the complete CyberSOC feature set including unlimited data retention (configurable), 1 TB+ daily ingestion, full AI Copilot suite with autonomous investigation, MSSP multi-tenancy support, advanced compliance frameworks (SOC 2, ISO 27001, GDPR), custom integrations via professional services, and 24/7 premium support with 1-hour critical issue SLA. Enterprise customers receive assigned technical account managers, architecture consultations, and early access to beta features. Contract terms are customizable with annual or multi-year commitments attracting significant discounts.

Government Tier: Specifically architected for federal, state, and local government agencies, the Government tier incorporates all Enterprise capabilities plus specialized features addressing public sector requirements. These include FedRAMP High authorization artifacts, IL5/IL6 compliance packages, CUI handling controls, government-specific reporting formats (FISMA, OMB), sovereign deployment options, and dedicated government compliance specialists. Pricing follows GSA Schedule structures where applicable, with specialized contract vehicles for direct agency procurement. The Government tier undergoes additional security review and includes enhanced audit logging meeting government accountability standards.

On-Premise Tier: Addressing air-gapped and highly regulated environments requiring complete data sovereignty, the On-Premise tier delivers full CyberSOC capability through customer-hosted infrastructure. This tier includes perpetual or subscription licensing options, offline license activation mechanisms, air-gapped update distribution via secure media, hardware-based license enforcement (USB dongle or server-bound), and optional professional services for deployment and maintenance. On-Premise pricing reflects the premium nature of isolated deployment including costs for customized delivery mechanisms, reduced economies of scale, and enhanced support requirements for self-managed infrastructure.

3. Feature Matrix by Tier

The following comprehensive feature matrix provides detailed visibility into capability availability across all six licensing tiers. This matrix serves as the definitive reference for sales engineering, solution architecture, and customer conversations regarding feature inclusion. Each row represents a distinct platform capability organized by functional category, while columns indicate tier availability through standardized symbols indicating full inclusion, partial availability with limitations, add-on purchase requirement, or exclusion from the tier. Product management should review and update this matrix quarterly to reflect roadmap deliveries and packaging decisions.

3.1 Core Platform Features

Feature	Free	Pro	Business	Enterprise	Gov	On-Prem
SIEM Core Engine	Basic	Full	Full	Full	Full	Full
Data Retention	7 days	30 days	90 days	Unlimited	Unlimited	Configurable
Daily Ingestion	1 GB	10 GB	100 GB	1 TB+	1 TB+	License-based
User Seats	5	25	100	Unlimited	Unlimited	License-based
Dashboard Views	Analyst Only	+Manager	+CISO	All	All	All
Alert Correlation	Rule-based	ML Basic	ML Advanced	AI Full	AI Full	AI Full
API Access	Read-only	Full REST	Full + WS	Full + Custom	Full + Gov	Full

Table 3.1: Core Platform Feature Matrix

3.2 Advanced Security Capabilities

Capability	Free	Pro	Business	Enterprise	Gov	On-Prem
UEBA Engine	-	Basic	Full	Full + AI	Full + AI	Full
XDR Integration	-	-	Endpoint	Full XDR	Full XDR	Full XDR
NDR/Network Detection	-	-	Basic	Full	Full	Full
SOAR Automation	-	50/mo	500/mo	Unlimited	Unlimited	Unlimited
Threat Intel Feeds	Open Source	Commercial	Premium	Custom	Gov-Specific	Configurable
Threat Hunting	Manual	Guided	Advanced	AI-Assisted	AI + Classified	AI-Assisted
DFIR Platform	-	Basic	Full	Full + Auto	Full + Auto	Full
Attack Surface Mgmt	-	-	External	Full	Full + Gov	Full

Table 3.2: Advanced Security Capabilities Matrix

3.3 AI and Intelligence Features

AI Feature	Free	Pro	Business	Enterprise	Gov	On-Prem
AI SOC Copilot	Suggest	Assist	Auto-Pilot	Autonomous	Autonomous	Auto-Pilot
Natural Language Query	Limited	Full	Full + Context	Advanced	Advanced	Full
Automated Investigation	-	Basic	Standard	Full Auto	Full Auto	Standard
Predictive Analytics	-	Trends	Full	Prescriptive	Prescriptive	Full
ML Detection Models	Pre-built	Custom (5)	Custom (25)	Unlimited	Unlimited	Custom (50)
Security Graph	View	Analyze	Full	AI Insights	AI + Gov Data	Full
Risk Engine	Score	Quantify	Predict	Simulate	Simulate	Predict

Table 3.3: AI and Intelligence Feature Matrix

3.4 Operations and Support Features

Feature	Free	Pro	Business	Enterprise	Gov	On-Prem
Support Channel	Forum	Email	Email + Phone	24/7 Premium	24/7 Dedicated	24/7 Premium
Response SLA	Best Effort	4 hours	2 hours	1 hour Critical	1 hour Critical	4 hours
Dedicated Manager	-	-	Success Mgr	Account Mgr	Gov Specialist	Success Mgr
Training Included	Documentation	Online	Online + Workshops	On-site	On-site + Clearinghouse	Workshop
Professional Services	-	-	Optional	Included Hours	Included + Gov	Required Add-on
SLA Uptime Guarantee	99.5%	99.9%	99.95%	99.99%	99.99%	Best Effort
Disaster Recovery	-	Backup	Active DR	Geo-redundant	Sovereign DR	Customer Managed
MSSP Multi-tenancy	-	-	Basic (5)	Full (Unlimited)	Full + Gov	Full (Licensed)

Table 3.4: Operations and Support Feature Matrix

3.5 Compliance and Governance

Compliance Area	Free	Pro	Business	Enterprise	Gov	On-Prem
Compliance Frameworks	Basic Mapping	SOC 2 Type I	SOC 2 Type II	Full Suite	FedRAMP + Full	Configurable
Audit Logging	Basic	Standard	Comprehensive	Tamper-proof	Gov Archive	Local Storage
Data Residency	Default Region	Choice of Region	Dedicated Tenancy	Sovereign Cloud	Required	Customer Controlled
Role-Based Access	3 Roles	5 Roles	10 Roles	Custom + ABAC	ABAC + Clearinghouse	Custom
Data Encryption	At Rest/Transit	+ Key Mgmt	+ BYOK	+ HSM	+ FIPS 140-2	+ HSM Option
Vulnerability Mgmt	Scan Reports	Prioritization	Remediation	Full Lifecycle	Full + FedRAMP	Full Lifecycle
Cloud Security	-	CSPM Basic	CSPM Full	CNAPP	CNAPP + Gov	CSPM Full
Identity Security	SSO Basic	IdP Integration	Full IAM	PAM + CIEM	PIV/CAC Support	Full IAM

Table 3.5: Compliance and Governance Matrix

4. Pricing Model Architecture

The CyberSOC pricing model employs a hybrid approach combining subscription-based recurring revenue with usage-based components for variable consumption. This architecture balances predictable budgeting for customers with fair value exchange for the platform provider. The model has been designed following extensive market analysis of competing SIEM/SOAR platforms, customer interviews spanning organizational sizes, and financial modeling ensuring unit economics sustainability across all customer segments and growth scenarios.

4.1 Base Subscription Pricing

Tier	Model	Base Price	Billing Cycle	Contract Min
Free	Perpetual Free	\$0	N/A	None
Professional	Per-Analyst/Month	\$149/user/mo	Monthly/Annual	5 Users
Business	Per-Analyst/Month	\$299/user/mo	Annual	25 Users
Enterprise	Platform + Consumption	\$50,000/yr base	Annual/Multi-year	100 Users
Government	Platform + Compliance	GSA Schedule	Annual/Multi-year	Agency Contract
On-Premise	Perpetual or Subscription	Custom Quote	Perpetual/3-yr sub	PO Required

Table 4.1: Base Subscription Pricing Structure

4.2 Usage-Based Pricing Components

Beyond base subscriptions, certain tiers incur usage-based charges for consumption that scales with operational intensity. This approach ensures customers pay proportionally to value received while avoiding over-provisioning costs typical of pure seat-based models. Usage metering occurs in real-time with granularity sufficient for cost allocation and showback reporting. Customers can set budget alerts and hard limits to prevent unexpected overage charges, with grace periods and burst allowances designed to accommodate legitimate security incident response scenarios without penalty.

Metric	Unit	Pro Rate	Business Rate	Enterprise Rate
Data Ingestion (overage)	per GB	\$0.50	\$0.30	\$0.15
Storage (extended retention)	per GB-month	\$0.20	\$0.12	\$0.08
SOAR Actions (overage)	per 1,000	\$10.00	\$5.00	\$2.00
API Calls (overage)	per 10,000	\$1.00	\$0.50	\$0.20

Metric	Unit	Pro Rate	Business Rate	Enterprise Rate
AI Copilot Queries	per 1,000	\$5.00	\$3.00	\$1.00
Threat Intel Enrichment	per 10,000	\$2.00	\$1.00	\$0.50
Report Generation	per 100	\$3.00	\$1.50	\$0.50
Long-term Retention (>1yr)	per GB-month	-	\$0.08	\$0.04

Table 4.2: Usage-Based Pricing Components

4.3 Volume Discount Structure

Enterprise and Government tiers benefit from volume-based discount structures that reduce effective per-unit costs as commitment increases. Discounts apply cumulatively across multiple dimensions including user count, data volume, contract duration, and multi-product bundling. The discount matrix below illustrates standard discount percentages; negotiated arrangements for strategic accounts may exceed these guidelines with appropriate executive approval. Discount tiers reset annually unless locked through multi-year agreements, providing flexibility for growing customers while protecting committed volumes from price increases.

Commitment Level	User Discount	Ingestion Discount	Multi-Year Bonus
Base (100-250 users)	0%	0%	0%
Silver (251-500 users)	10%	8%	5% (2-yr), 10% (3-yr)
Gold (501-1000 users)	18%	15%	8% (2-yr), 15% (3-yr)
Platinum (1001-2500 users)	25%	22%	12% (2-yr), 20% (3-yr)
Diamond (2500+ users)	35%	30%	15% (2-yr), 25% (3-yr)
Strategic (Custom)	Negotiated	Negotiated	Up to 30%

Table 4.3: Volume Discount Structure

5. License Management System

The License Management System (LMS) forms the technical backbone of CyberSOC's billing architecture, responsible for entitlement verification, enforcement, metering, and compliance monitoring. The LMS has been architected as a microservice within the CyberSOC platform ecosystem, exposing well-defined APIs for integration with admin interfaces, payment processors, and external systems. The design prioritizes high availability given that licensing failures can impact platform availability for all customers, implementing graceful degradation modes that maintain functionality during transient outages while preventing systematic abuse of licensing boundaries.

5.1 System Architecture

The LMS comprises several interconnected components each serving distinct functions within the licensing lifecycle. The Entitlement Store maintains the authoritative record of customer permissions, implemented as an eventually consistent distributed database optimized for read-heavy workload patterns typical of license verification requests. The Policy Engine evaluates entitlement rules against current state, supporting complex logic including time-bound trials, graduated features, and conditional access based on compliance status. The Metering Service tracks usage against entitled limits with configurable granularity and retention periods supporting both real-time enforcement and historical billing reconciliation.

The Enforcement Agent runs as a trusted component within customer deployments, making local licensing decisions without requiring constant connectivity to central services. This architecture is particularly important for On-Premise deployments where internet access may be restricted. Agents periodically synchronize with the central LMS to receive updated entitlements and report metered usage, caching credentials and permissions locally with tamper-evident storage. The Audit Logger maintains an immutable chain of all licensing decisions and state changes, supporting compliance investigations and dispute resolution with cryptographic integrity verification.

Component	Function	Technology	SLA
Entitlement Store	Authoritative permission record	PostgreSQL + Redis Cache	99.99%
Policy Engine	Rule evaluation and decisions	Go microservice + OPA	99.95%
Metering Service	Usage tracking and aggregation	Apache Kafka + ClickHouse	99.9%
Enforcement Agent	Local decision making	Rust binary (embedded)	Always Available
Audit Logger	Immutable decision log	Append-only S3 + signing	99.99%

Component	Function	Technology	SLA
Sync Service	Agent coordination	gRPC + mTLS	99.9%
Admin API	Management interface	REST + OpenAPI 3.0	99.95%

Table 5.1: License Management System Components

5.2 License Key Structure

CyberSOC employs cryptographic license keys that encode entitlement information in a tamper-evident format. Each license key contains signed claims specifying permitted capabilities, limitations, validity period, and deployment constraints. The key structure follows JWT-like conventions with platform-specific extensions for licensing semantics. Keys are issued by the central licensing authority using ECDSA P-256 signatures, verifiable by enforcement agents without online connectivity. This design enables offline operation for On-Premise deployments while maintaining cryptographic proof of authenticity and integrity.

License keys include both public claims visible to administrators for troubleshooting and private claims encrypted for enforcement only. Public claims include tenant identifier, tier level, feature flags, user limits, and expiration timestamp. Private claims contain machine-binding hashes, anti-circumvention tokens, and usage counters for metered features. The hybrid structure balances transparency for operational effectiveness with security measures preventing license manipulation or sharing beyond authorized scope.

Claim Type	Field	Description	Example
Public	tid	Tenant UUID	550e8400-e29b...
Public	tier	License tier	enterprise
Public	features	Enabled feature list	siem,xdr,soar,ai
Public	max_users	Seat limit	500
Public	exp	Expiration epoch	1735689600
Private	hw_hash	Hardware binding	sha256:abc123...
Private	usage_sig	Usage counter signature	integrity token
Private	node_limit	Deployment node limit	8

Table 5.2: License Key Claim Structure

6. Subscription Lifecycle Management

Subscription Lifecycle Management encompasses all processes governing customer relationships from initial acquisition through renewal, expansion, churn, and potential reactivation. The CyberSOC lifecycle model has been designed to maximize customer lifetime value while ensuring frictionless experience for legitimate use cases. Automated workflows handle routine transitions while flagging edge cases for human review. Integration with CRM systems, revenue recognition platforms, and customer success tools ensures consistent experience across all touchpoints while maintaining accurate financial records for reporting purposes.

6.1 Customer Journey Stages

Stage	Duration	Key Activities	System Actions
Trial/Free	14-30 days	Evaluation, POC, integration testing	Feature-gated, usage capped
Onboarding	First 30 days	Configuration, training, data migration	Success manager assignment
Active Normal	Ongoing	Production usage, optimization	Metering, health checks
Renewal Window	90 days pre-expiry	Review, negotiation, contract update	Renewal alerts, quoting
Expansion	Triggered	Upsell, tier upgrade, add-on purchase	Revised billing adjustment
At-Risk	Detected	Engagement decline, issues flagged	Intervention triggers, outreach
Churned	Post-cancellation	Offboarding, data export, feedback	Access downgrade, retention
Reactivated	New contract	Return path, win-back incentives	History restoration option

Table 6.1: Subscription Lifecycle Stages

6.2 Automated Workflow Triggers

The lifecycle management system implements automated triggers responding to significant events and threshold crossings. These triggers initiate predefined workflows ensuring timely action without requiring manual monitoring of all customer accounts. Trigger conditions have been carefully tuned to balance sensitivity (catching genuine issues early) against false positive rate (avoiding alert fatigue for customer success teams). Machine learning models continuously refine trigger thresholds based on historical outcomes, improving prediction accuracy for churn risk and expansion opportunity identification over time.

Trigger Event	Condition	Action Initiated	Owner
Trial Expiring	3 days remaining	Extension offer + conversion call	Sales Development

Trigger Event	Condition	Action Initiated	Owner
Usage Threshold	>80% of limit	Upgrade recommendation	Customer Success
Payment Failure	Retry exhausted	Grace period + suspension notice	Billing Operations
Low Engagement	<20% MAU for 2 weeks	Health check outreach	Customer Success
Support Escalation	3+ severity 1 tickets	Technical account review	Technical Account Mgr
Contract Expiry	90 days before	Renewal preparation start	Account Executive
Expansion Signal	Job change / funding	Opportunity qualification	Sales Representative

Table 6.2: Automated Workflow Triggers

6.3 Grace Period and Suspension Policies

Recognizing that security operations cannot tolerate abrupt service termination, CyberSOC implements graded response to payment and compliance issues. Upon payment failure, customers enter a 14-day grace period during which all services continue normally while collection efforts proceed. If unresolved, soft restriction mode limits non-essential features (AI copilot, advanced analytics) while preserving core security functionality for an additional 14 days. Hard suspension, terminating all access except audit log export, occurs only after 30 days of non-payment. This staged approach protects both customer security posture and vendor revenue while maintaining ethical business practices appropriate for critical infrastructure software.

7. Enterprise & Government Features

Enterprise and Government tiers represent the pinnacle of CyberSOC capability, incorporating advanced features necessary for large-scale security operations with demanding compliance requirements. These tiers go beyond incremental feature additions to provide fundamentally different architectural capabilities including multi-tenant isolation for MSSP operations, sovereign deployment options for data residency compliance, and specialized workflows addressing government-specific mandates. Investment in these tiers reflects their strategic importance, typically representing majority of platform revenue despite smaller customer counts compared to lower tiers.

7.1 MSSP Multi-Tenancy (Enterprise+)

Managed Security Service Providers (MSSPs) require ability to serve multiple customer tenants from single CyberSOC deployment while maintaining complete logical isolation between tenant data, configurations, and user access. The Enterprise tier includes basic MSSP support for up to 5 tenants, while full Enterprise and Government tiers remove tenant count limitations. Each tenant receives independent data retention policies, alert thresholds, dashboard configurations, and compliance reporting. Tenant administrators manage their environment within guardrails defined by the MSSP, preventing configuration drift while enabling operational autonomy appropriate for delegated security management relationships.

MSSP Capability	Business	Enterprise	Government
Maximum Tenants	5	Unlimited	Unlimited
Tenant Isolation	Logical	Logical + Crypto	Logical + Crypto + Air-gap
White-label Options	Basic	Full UI + Reports	Full + Gov Branding
Tenant Portal	Shared	Dedicated per Tenant	Dedicated + Agency Portal
Cross-tenant Search	-	With Authorization	With Clearance Check
Resource Pooling	Static	Dynamic Quotas	Dynamic + Priority
Tenant Billing	Included	Chargeback Support	Fund Code Tracking
SLA per Tenant	Global	Per-tenant Configurable	Per-tenant + Mission Class

Table 7.1: MSSP Multi-Tenancy Feature Comparison

7.2 Government-Specific Capabilities

The Government tier incorporates specialized capabilities addressing unique requirements of public sector security operations. These extend beyond standard enterprise features to include compliance artifacts for authorization frameworks commonly required in government procurement, handling controls for classified and sensitive information, and deployment architectures meeting sovereignty and air-gap requirements. Government customers work with dedicated compliance specialists who understand agency-specific requirements and can guide authorization package development, security assessment coordination, and continuous monitoring configuration to maintain authority to operate (ATO) status throughout the engagement lifecycle.

Capability	Description	Standards
FedRAMP Ready	Package for authorization	FedRAMP High Baseline
IL5/IL6 Controls	Impact level handling	DoD Cloud Computing SRG
CUI Marking	Controlled unclassified info	NIST SP 800-171 / CUI Registry
FISMA Reporting	Agency compliance reports	OMB Circular A-123, FISMA Act
PIV/CAC Auth	Government smartcard login	HSPD-12 / NIST SP 800-116
Sovereign Deployment	Data residency guarantee	FedRAMP + Agency Requirements
Classified Processing	Optional SCI enclave	ICD 503 / CNSSI 1253
Continuous Diagnostics	CDM integration ready	CDM Program Requirements

Table 7.2: Government-Specific Capabilities

8. On-Premise Licensing Model

The On-Premise licensing model addresses scenarios requiring complete infrastructure control due to security policy, regulatory mandate, or network topology constraints preventing cloud connectivity. Common drivers include classified processing environments, air-gapped operational technology networks, data sovereignty requirements mandating physical jurisdictional control, and organizational policies prohibiting SaaS consumption for security tooling. The On-Premise tier delivers equivalent functional capability to cloud deployments through alternative delivery and licensing mechanisms adapted to disconnected operation.

8.1 Licensing Mechanisms for Air-Gapped Environments

Air-gapped environments present unique challenges for license enforcement typically reliant on cloud connectivity for validation. CyberSOC addresses this through multiple complementary mechanisms providing appropriate assurance levels for different threat models. The primary mechanism uses cryptographically signed license files encoding entitlements with validity periods, distributed through secure media transfer processes. For higher assurance requirements, hardware dongles (USB tokens) provide tamper-resistant credential storage with server binding preventing license migration beyond authorized deployments. Server-bound licenses utilize hardware fingerprinting to tie entitlements to specific physical infrastructure, balancing usability against unauthorized copying prevention.

Mechanism	Security Level	Connectivity	Use Case
Signed License File	Standard	Periodic sync	Typical on-premise
Hardware Dongle	High	Initial + renewal	High-value deployments
Server Binding	Medium-High	Initial only	Stable infrastructure
Air-Gap Transfer Media	Varies	Physical transfer	Classified networks
Offline Activation	Standard	Phone/email one-time	Limited connectivity
Satellite Sync Server	High	Internal relay	Large air-gapped orgs

Table 8.1: On-Premise Licensing Mechanisms

8.2 Update Distribution and Maintenance

Software updates for On-Premise deployments follow secure supply chain practices ensuring integrity and authenticity of all delivered components. Updates are packaged with cryptographic signatures verifiable by the target deployment before installation. For

air-gapped environments, updates distribute through designated transfer media following procedures defined in customer security plans. The update manifest includes hash verification, dependency specifications, rollback instructions, and security bulletin references. Customers with active maintenance subscriptions receive update notifications and media on established schedules (quarterly for patches, annual for feature releases) with expedited delivery available for critical security patches affecting deployed versions.

Maintenance tiers for On-Premise deployments range from self-service (download and install independently) through assisted (vendor-guided remote sessions during maintenance windows) to fully managed (vendor personnel on-site for major updates). Higher maintenance tiers include additional benefits such as customized patch scheduling aligned with change windows, backward compatibility guarantees for customer integrations, and dedicated engineering liaison for issue resolution. Maintenance pricing scales with deployment complexity and service level expectations, typically ranging from 18-25% of license fee annually for standard maintenance coverage.

9. Billing Engine Design

The Billing Engine translates licensing entitlements and usage measurements into accurate, timely invoices while maintaining audit trails sufficient for financial compliance and dispute resolution. The engine has been designed for integration with multiple payment processors, tax calculation services, and ERP systems commonly used by enterprise customers. Architecture supports both advance billing (subscription fees billed at period start) and arrears billing (usage overages billed based on actual consumption), with configurable billing cycle boundaries accommodating diverse customer fiscal calendar requirements.

9.1 Billing Pipeline Architecture

The billing pipeline operates as a sequence of processing stages transforming raw usage data into formatted invoices. Collection stage aggregates metered usage from distributed sources, normalizing timestamps and handling late-arriving data through configurable watermark delays. Rating stage applies pricing rules including tier determination, discount calculation, and promotional credit application. Tax stage determines applicable taxation based on customer location, product taxonomy, and exemption certificates on file. Invoice generation stage produces formatted documents with line item detail, supporting multiple formats (PDF, EDI, API payload) for delivery through configured channels. Finally, payment processing stage manages collection attempts, dunning sequences, and receivables accounting integration.

Stage	Input	Processing	Output
Collection	Raw meter events	Aggregation, deduplication, normalization	Usage records
Rating	Usage records	Price lookup, tier calc, discounts	Rated transactions
Tax	Rated transactions	Jurisdiction lookup, exemption check	Tax-inclusive charges
Invoice Gen	Taxed charges	Template rendering, formatting	Invoice documents
Delivery	Invoice documents	Channel routing, notification	Customer receipt
Payment	Delivered invoices	Collection attempt, reconciliation	Payment records
Recognition	Payment records	Revenue scheduling, GL posting	Financial entries

Table 9.1: Billing Pipeline Stages

9.2 Payment Processing Integration

The billing engine integrates with multiple payment processors to accommodate regional preferences and enterprise procurement requirements. Credit card processing

supports Visa, Mastercard, American Express, and JCB through PCI-DSS compliant tokenization. ACH/wire transfer handling accommodates enterprise customers preferring bank-to-bank settlement with automated reconciliation matching payments to invoices based on reference numbers. Purchase order processing supports customers requiring formal PO-based workflows with approval routing and budget checking before invoice issuance. For international customers, local payment methods (SEPA, BACS, PayPal) expand accessibility while localized tax handling addresses VAT/GST requirements across jurisdictions.

Method	Supported Tiers	Settlement	Fees
Credit Card	Pro, Business	Real-time (2-3 day)	2.9% + \$0.30
ACH Transfer	All tiers	2-5 business days	0.5% cap \$5
Wire Transfer	Enterprise, Gov	1-3 business days	Customer pays fee
Purchase Order	Business+	Net 30/60 terms	None (credit check)
Cryptocurrency	On request	Variable	Market rate

Table 9.2: Supported Payment Methods

10. Integration with Admin Interface

The Billing and Licensing Architecture integrates deeply with the CyberSOC Admin Interface, providing unified administrative control over all commercial and operational aspects of the platform. Administrators with appropriate permissions can manage customer accounts, adjust entitlements, investigate billing disputes, configure pricing rules, and monitor system health through consistent UI patterns established across the admin interface. This section details the specific administrative capabilities exposed through the admin interface for billing and licensing operations, organized by functional area and role requirement.

10.1 Administrative Capabilities Matrix

Capability	Super Admin	Org Admin	Billing Admin	ReadOnly
Create/Edit Licenses	Full	Own Org	Own Org	View
Adjust Entitlements	Full	Within Limits	Credit Adjust	View
Process Refunds	Full	-	With Approval	-
Configure Pricing Rules	Full	-	-	-
View Revenue Dashboards	Full	Own Org	Full	Aggregate
Manage Payment Methods	Full	Own Org	Full	-
Export Billing Data	Full	Own Org	Full	Redacted
Audit Log Access	Full	Own Org	Billing Scope	Summary
Tenant Management (MSSP)	Full	Own Tenants	-	View Own

Table 10.1: Administrative Capabilities by Role

10.2 Admin Dashboard Components

The Admin Interface exposes dedicated dashboard components for billing and licensing oversight. The Revenue Overview component displays MRR (Monthly Recurring Revenue), ARR (Annual Recurring Revenue), growth trends, and cohort analysis broken down by tier, geography, and channel. The License Inventory component shows active license counts by tier, expiration timeline, utilization rates, and renewal probability scores. The Usage Analytics component highlights customers approaching limits, unusual consumption patterns, and opportunities for upsell based on feature adoption. The Collections Dashboard tracks aging receivables, payment failure rates, dunning progress, and cash flow projections enabling proactive revenue operations management.

Component	Data Sources	Refresh	Export
Revenue Overview	Billing + CRM	Hourly	CSV, PDF, API
License Inventory	LMS Entitlement Store	Real-time	CSV, JSON
Usage Analytics	Metering Service	15 minutes	CSV, Dashboard embed
Collections Status	Payment Processor	Real-time	PDF Report
Customer Health	CS Platform + Usage	Daily	Scorecard PDF
Compliance Monitor	Audit Logger	Hourly	Attestation Package

Table 10.2: Admin Dashboard Components

11. Compliance & Audit Framework

The billing and licensing system operates within a comprehensive compliance framework addressing financial regulations, data protection requirements, and industry-specific standards relevant to cybersecurity platforms serving global customers. This framework establishes controls, documentation requirements, and audit mechanisms demonstrating compliance to internal stakeholders, external auditors, and regulatory bodies. Compliance considerations have been integrated into system design rather than applied as afterthoughts, ensuring that auditability and control effectiveness are inherent properties of the architecture rather than superficial overlays.

11.1 Applicable Standards and Regulations

Standard	Scope	Key Requirements	Status
SOC 2 Type II	Financial controls	Access controls, encryption, availability	Certified
GDPR	EU data protection	Data minimization, consent, right to deletion	Compliant
CCPA/CPRA	California privacy	Disclosure, opt-out, data portability	Compliant
PCI-DSS	Cardholder data	Tokenization, network segmentation	Level 1 Compliant
SOX	Financial reporting	Segregation of duties, audit trail	Aligned
ISO 27001	Information security	ISMS, risk treatment, continual improvement	Certified
FedRAMP	US Government	Security controls, continuous monitoring	In Progress
GDPR Art. 82-84	Breach notification	72-hour notification, documentation	Implemented

Table 11.1: Applicable Compliance Standards

11.2 Audit Logging and Evidence Preservation

All billing and licensing operations generate immutable audit logs capturing actor identity, action taken, timestamp, affected resources, and outcome. Logs append to tamper-evident storage with cryptographic chaining ensuring detection of any retrospective modification. Retention periods align with regulatory requirements (minimum 7 years for financial records, longer where contractually specified). Export mechanisms produce evidence packages in formats acceptable to external auditors, with attestation documentation describing collection methodology and integrity verification procedures. Regular access reviews confirm continued appropriateness of personnel with billing system access, with immediate revocation upon role change or termination.

11.3 Data Protection Measures

Customer billing data receives heightened protection reflecting sensitivity of financial information and privacy obligations. Data classification marks billing records as Confidential with handling requirements including encryption at rest (AES-256) and in transit (TLS 1.3), access logging, and need-to-know access controls. Data minimization principles limit collection to information necessary for billing purposes, with separate consent required for secondary uses such as analytics or marketing. Customer rights requests access, correction, deletion, and portability process within statutory timeframes (typically 30 days) through documented procedures with escalation paths for complex cases.

12. Implementation Roadmap

Implementation of the Billing and Licensing Architecture follows a phased approach managing risk while delivering incremental value. Each phase builds upon previous deliverables, establishing foundations before adding complexity. The roadmap assumes parallel workstreams for backend infrastructure and admin interface development, with integration points clearly defined at phase boundaries. Timeline estimates assume adequately staffed cross-functional teams with necessary domain expertise; adjustments may be required based on organizational context and dependencies on external systems or third-party integrations.

12.1 Phase Timeline

Phase	Duration	Deliverables	Dependencies
Phase 1: Foundation	Weeks 1-6	LMS core, basic entitlement store, license key format	Infrastructure ready
Phase 2: Core Billing	Weeks 7-12	Subscription management, invoicing, payment integration	Phase 1 complete
Phase 3: Usage Metering	Weeks 13-18	Metering pipeline, usage rating, overage billing	Phase 2 complete
Phase 4: Admin Interface	Weeks 19-24	Admin dashboards, license management UI	Phase 3 APIs
Phase 5: Advanced Tiers	Weeks 25-32	Enterprise features, MSSP tenancy, Gov capabilities	Phase 4 complete
Phase 6: On-Premise	Weeks 33-40	Offline licensing, air-gap support, secure updates	Phase 5 stable
Phase 7: Compliance	Weeks 41-48	Audit automation, evidence packages, certification	Phase 6 stable
Phase 8: Optimization	Ongoing	Performance tuning, ML pricing, fraud detection	Production learnings

Table 12.1: Implementation Phase Timeline

12.2 Success Criteria and Milestones

Each phase includes defined success criteria that must be met before progression to subsequent phases. Gate reviews involving stakeholders from engineering, product, finance, and legal assess completion against criteria, identify risks, and approve advancement. Criteria combine functional completeness (all specified features working correctly), non-functional requirements (performance, availability, security benchmarks met), and operational readiness (documentation complete, support trained, runbooks validated). Phase gates prevent accumulation of technical debt and ensure organizational readiness for capabilities being delivered.

Milestone	Target Date	Criteria	Sign-off
LMS Alpha	Week 4	Core entitlement flow working end-to-end	Engineering Lead

Milestone	Target Date	Criteria	Sign-off
Billing Beta	Week 10	Can invoice and collect payment for test customers	Product + Finance
GA Release	Week 24	All tiers available for new customer signups	Executive Sponsor
Enterprise GA	Week 32	MSSP + Gov features certified for production	Security + Legal
On-Premise GA	Week 40	Air-gap licensing validated with pilot customer	Field Engineering
Compliance Cert	Week 52	SOC 2 + ISO 27001 surveillance complete	Compliance Officer

Table 12.2: Key Milestones and Success Criteria

This Billing and Licensing Architecture establishes the foundation for sustainable commercial operations of the CyberSOC Platform across diverse customer segments and deployment models. The six-tier structure provides clear value propositions for each target audience while the technical implementation ensures scalable, compliant, and user-friendly license management. Integration with the Admin Interface creates unified operational control, reducing overhead while improving visibility into commercial health. Regular review and evolution of this architecture will ensure continued alignment with market dynamics, customer needs, and regulatory requirements as the CyberSOC Platform grows and matures in the marketplace.